

From Likes To Leases: What Facebook Teaches Us About the Housing Market

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The Problem

Nowadays, finding an apartment for rent in Israel usually starts by opening your facebook feed. The rental market in Israel, which has been expanding in recent years¹, is mostly inhabited by the younger generations². This demographic trend can partially explain the extensive traffic in the housing-related facebook groups. These Facebook groups number in the tens of thousands of members, with dozens of new rental ads posted daily, making them a central meeting point for renters and landlords.³.

We will focus on the renting market in Jerusalem. Since not all apartments were created equal, we will try to uncover **what makes certain apartment posts more attractive**. Why are some posts so appealing, gaining substantial engagment online, while others stay in the dark. We will focus both on regular apartment rentals and on roommate-based shared living.

Data

We collected data from two facebook groups - "[Apartments for Rent in Jerusalem](#)", with 96K members, and "[Roommate Apartments In Jerusalem](#)", with 34.7K members.

We collected 3000 posts from each group, with a total of 6000 posts weighing approximately 8 MB of data. After filtering duplicates by content (2 posts with the same text posted at different times will be considered 1), we have 4598 unique posts. Almost all (99%) of these posts were posted between June 2024 and May 2025.

We also attempted to collect associated images, but encountered substantial technical difficulties.

Each record contains the following information:

- Post Textual content (The post itself)
- Amount of photos in the post
- Link to post
- Amount of reactions received
- Amount of comments received
- Author information - name, Facebook profile
- Timestamp
- Group posted in

To collect the data, we built a crawler to fetch post links, automated the ESUIT exporter to download them as JSON, and merged everything into a single dataframe (CSV). Additionally, we used OverPassTurbo to get information from OpenStreetMap about streets and neighborhoods in Jerusalem.

Methodology

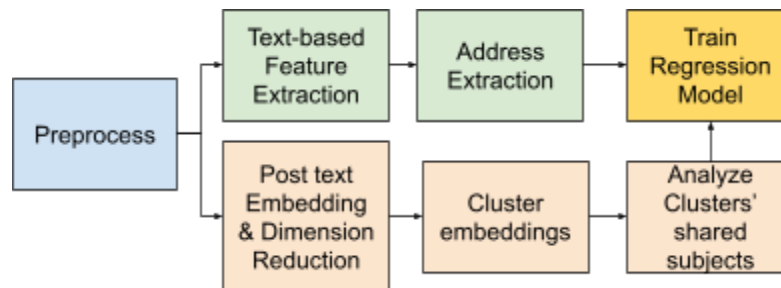
Since our goal is to better understand the latent mechanism around **engagement** in these particular circumstances, we want to maintain **simplicity** and **interpretability** for the model. We acknowledge the fact that it is possible to train some deep neural network to effectively predict the engagement, but we would thus **lose our ability to interpret** the model.

¹ [Buyitinisrael article](#)

² [JerusalemPolicyCenter](#)

³ [Acol.co.il](#)

Therefore, we chose a simpler method that involves more of the methods we've learned during the course. We believed that this workflow could balance interpretation and prediction:



1. Preprocessing - since the crawler's output was a directory full of JSON files., our first task was to create a dataframe from these files. Additionally, we made the following edits:
 - Removed irrelevant features from the crawling process
 - Removed duplicates with regard to post content - meaning if two posts had identical text, we deleted one instance.
 - Removed posts of people searching **for** apartments - since our goal is to understand the engagement of posts that **offer** apartments for rent, these searching posts are irrelevant. Regex was a helpful tool in identifying such posts.
2. Feature extraction - To improve our dataset with structured information, we developed a dedicated feature extraction class, implemented in `extract_features.py`. This module systematically parses unstructured Hebrew text from Facebook apartment listings to identify key attributes. We engineered binary features such as `search_post`, `porch`, `washing_machine`, `dryer`, `is_sublet`, `is_shabbat_kosher`, and `no_brokerage`, using regular expressions and domain-specific linguistic patterns we discovered along the way. Additionally, we extracted numerical and categorical fields, including `num_rooms` (with support for digit and Hebrew word formats, as well as fractional values like "חצי"), `rent_price` (with outlier filtering), and `contract_month`.
3. Address Extraction -Address extraction was tricky since posts often mention multiple or vague locations. We combined Regex (for exact matches) with fuzzy matching (for partial ones) to identify streets and neighborhoods, then refined rule-based criteria for higher precision (e.g., preferring neighborhood over street names). Finally, we mapped the extracted addresses onto Jerusalem's geographic layers to link each post to its location.
4. Embedding - In order to apply representational learning, we embedded the textual content of the posts using two techniques:
 - a. Using the Hebrew transformer-based language model "alephBERT", and applying mean pooling, we embedded each post into a vector of size 768.
 - b. Using the "heBERT-sentiment-analysis" model (another Hebrew variant of BERT), we analyzed the sentiment of each post, which received both a sentiment score and confidence.

For interpretability purposes, we did not feed the embedding directly into an RNN network, nor into the regression model. Instead, we chose to further process it in order to maintain our understanding of the semantic features.
5. Clustering - To uncover latent categories of apartment posts, we applied unsupervised clustering on the text embeddings. We first reduced the **768**-dimensional alephBERT embeddings using **PCA** to 50 dimensions for computational tractability and noise reduction. We then ran **KMeans** on the reduced vectors, using silhouette and Davies-Bouldin scores to select the number of clusters. This produced high-level groupings of semantically similar posts without supervision, which we later interpreted using textual analysis.
6. Cluster Textual Analysis - To interpret the resulting clusters, we used **TF-IDF** to extract the most distinctive words in each cluster. We further used Chi-squared feature selection to identify words that were statistically overrepresented in each group. Combining both methods allowed us to manually label clusters with semantic themes, such as "Religious female roommates", "Urgent reposts", or "Institutional housing".
7. Model Training - We used two simple linear models. Our baseline was plain OLS (`BaselineFinal_RAWONLY`) using the post's month and group indicator as parameters. We also trained a Ridge regression on the engineered tabular features, tuning α on the dev split and refitting on train+dev for the final test evaluation. For Ridge we evaluated $\alpha \in \{0.01, 0.1, 1, 10, 100\}$ on the dev split, selected the best model by dev RMSE, broke ties by MAE, then refit the chosen α on train+dev and evaluated once on the held-out test.

Evaluation

Evaluation criteria

1. Geolocating posts was challenging, especially when multiple locations were mentioned (e.g., “I live on Radak Street, two minutes from Gaza Road in Rehavia, and a 10-minute bus to Givat Ram”). To minimize errors, we prioritized precision over coverage—better to have fewer matches than incorrect ones. We tested different matching rules across five iterations, each time manually validating 100 posts. In the end, we achieved about 80% precision while covering 83% of the data.
2. Cluster Evaluation - We used standard metrics for unsupervised learning: silhouette score and Davies-Bouldin index. Both scores suggested weak cluster separation, which is expected when clustering on high-dimensional short text. However, manual inspection of cluster content via TF-IDF and χ^2 terms revealed coherent semantic themes, confirming the clusters were meaningful and useful as an additional categorical feature.
3. Model Evaluation - we use a standard Split: 60% train, 20% dev, 20% test, fixed random seed.
Selection: tune on dev using RMSE as primary, break ties by MAE.
Final: refit the chosen hyperparameters on train+dev, evaluate once on the held-out test.
Reporting: MAE, RMSE, R^2 , plus an actual-vs-predicted plot.
Main Success Metrics: The test improves over the baseline on RMSE and MAE, with higher R^2 . That is, we are able to meaningfully generalize using the engineered features (textual & semantic).
4. Baseline Model - Our baseline is plain ordinary least squares (`BaselineFinal_RAWONLY`). It uses the timestamp of publication and the group indicator. The model is fit on the train split and evaluated on dev/test.

Setup

As mentioned in previous parts, we first preprocessed the folder scheme into a dataframe, and performed several edits on the entire data. Then, we processed the data, again in its entirety, to produce textual, geographic and semantic features.

For training purposes, we split the data into 60% train, 20% dev, 20% test, segments, and fitted the regression models using the train segment.

Acquiring the ground-truth labels was easy, once we’ve scraped the posts. Since we were trying to predict engagement, we already had the metrics for engagement in each post.

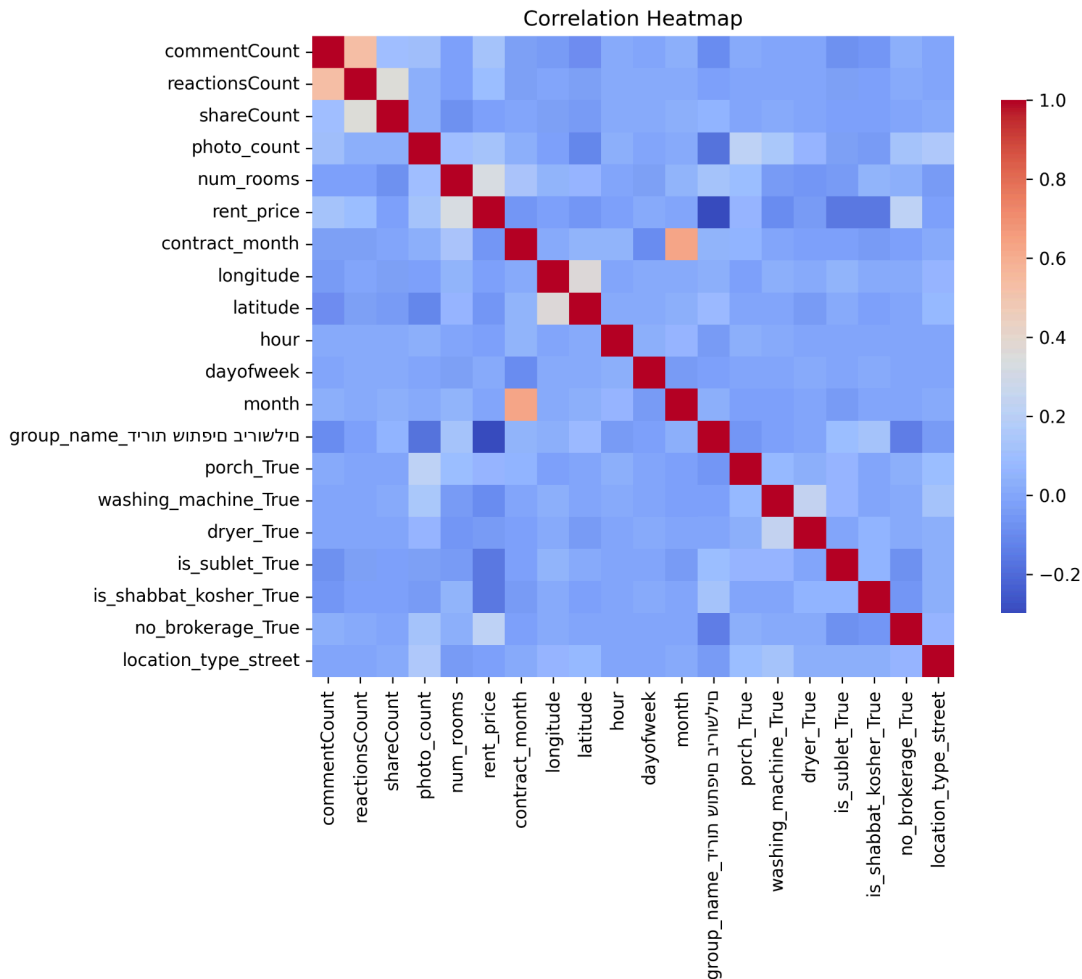
Avoiding bias was hard, since we a-priori did not know what the post is “talking about”, and therefore prone to outliers and irrelevant data. One method we used, mentioned earlier, was removing posts that are meant to **find** an apartment rather than renting one.

Engagement definition: we focus on reactions and comments and exclude share count, which in our setting mainly reflects coordination to boost visibility rather than authentic engagement.

Results & Visualization

Exploratory Data Analysis:

After the textual & geographic feature extraction, we can look at the correlations between pairs of features in the dataset:



We can see that no strong correlation exists between any pair, besides the contract start month and the post month. In particular, we do not see any strong linear correlation between the labels (commentCount and reactionsCount), and the rest of the data.

Data Geolocating

The following [MAP](#) shows the posts' geolocation, along with some filtered layers, which focus on price/Sublet/Kosherness.

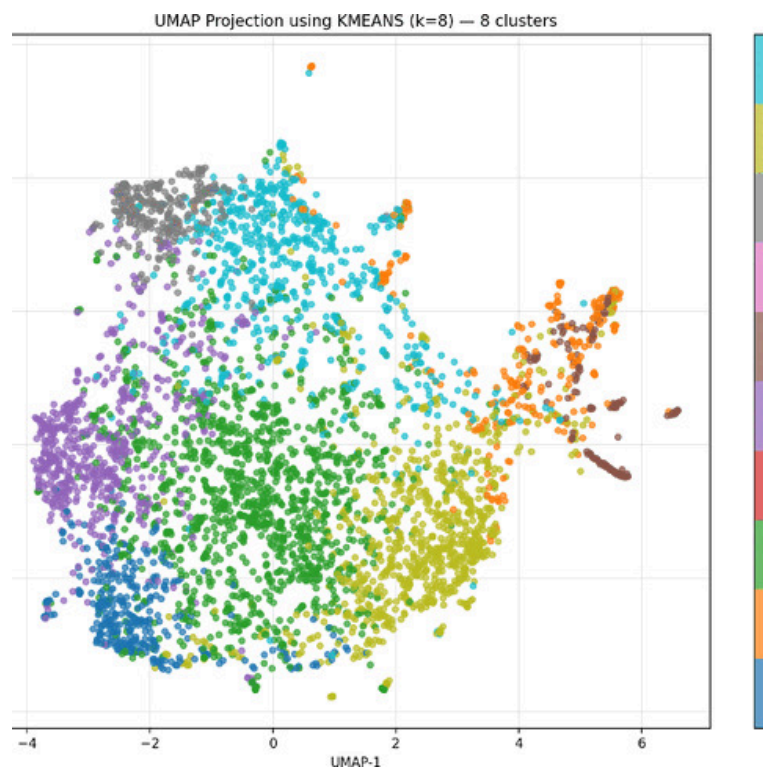
After we mapped all posts' geolocations and overlaid layers for Arab and Government/Industrial neighborhoods. This helped us see the boundaries of the Facebook rental market and which demographic groups it reaches. It's clear that Arab and Haredi areas have far fewer posts, suggesting they rely less on Hebrew-speaking Facebook groups for housing.

We also saw that both post frequency and apartment prices—reflecting demand and supply—peak around Rechavia, Nachlaot, and the city center, the city's three most popular neighborhoods. Finally, kosher apartments appear mainly in the southern areas, like Old Katamon and the Greek & German Colonies

Cluster Analysis

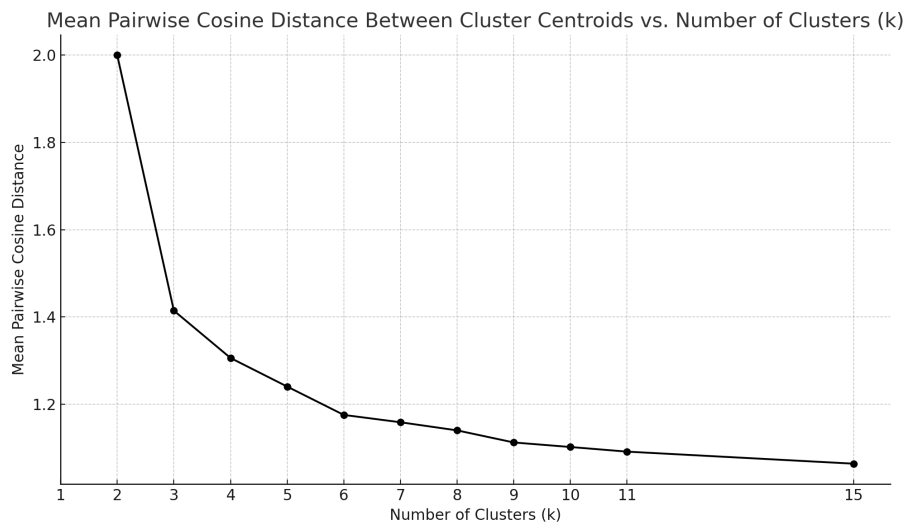
To better understand engagement patterns, we applied clustering to the embedded vectors representing each post. As shown in the animation, the data is inherently dense and lacks clear separations, especially at low values of k . This is reflected in poor internal validation metrics for small k , and negligible visual differences between $k = 7$ and $k = 8$. Because of this, we shifted our focus from geometric structure to

semantic meaning. We evaluated each clustering using TF-IDF and χ^2 scores to extract the most meaningful clusters in terms of language, context, and engagement. This approach allows us to interpret cluster content despite the ambiguous structure of the embedding space. Graph of the selected Clusters after PCA:



And a GIF of [the clustering process](#), (added as a link for technical reasons).

To support this shift, we also measured the average pairwise cosine distance between cluster centroids across different k values. As shown in the plot, the structural separation decreases steadily and flattens around $k = 8$, indicating diminishing returns from adding more clusters. This confirms our decision to prioritize semantic interpretability.



Below is a summary of the thematic meaning we assigned to each cluster at $k = 8$, based on TF-IDF and χ^2 term analysis:

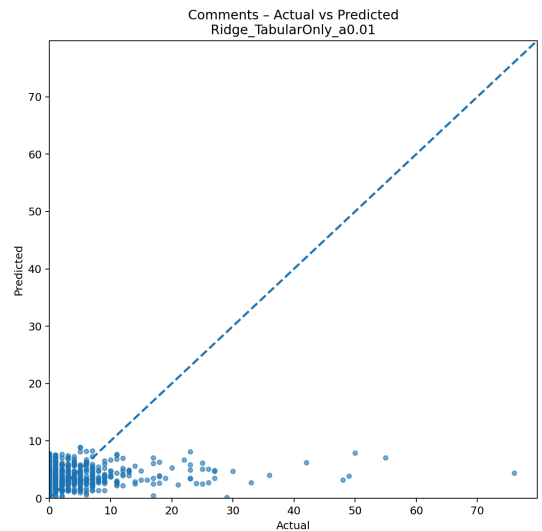
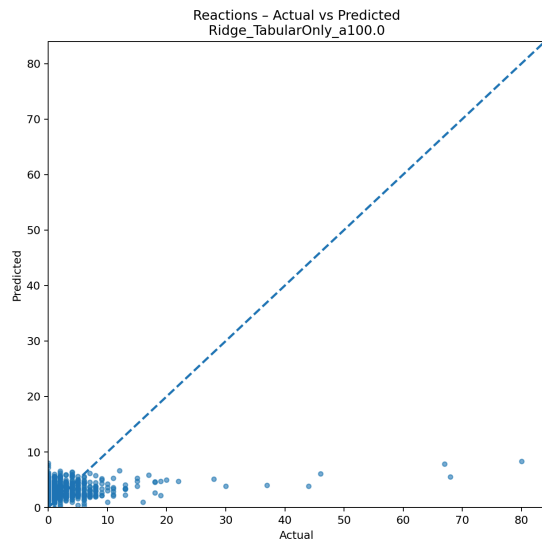
Cluster ID	Cluster Size (Relative Size%)	Semantic Category	Description
0	369 (8%)	"Kosher" Apartment Posts	Keywords like: "כשרות", "שבת", "סוכה".
1	284(6%)	English posts	Posts written in English, e.g., "available", "floor", "NIS"
2	1332(29%)	Coordination & Scheduling	Terms like "המיקום", "לתאם", related to arranging visits
3	643(15%)	Female Flatmates	Words like "שותפה", "משמחות", indicating shared housing
4	193(4%)	Broker / URL Spam	Names and domains like "cohen", "ben", "facebook", "com"
5	283(6%)	Student-Oriented	Mentions of "מדעי", "בצלאל", areas near central campuses
6	858(18%)	Family-Oriented	Words like "שירותים", "סורגים", "ילדים"; suitable for families
7	636(14%)	Personal Intro / Search	Terms like "נשמח", "מחפש", "שמי"; people seeking apartments

Model Analysis:

Enhanced Model result metrics:

Target	Best model	Model MAE (baseline MAE)	Model RMSE (baseline RMSE)	Model R ² (baseline R ²)
reactionsCount	Ridge_TabularOnly_a100.0	2.807 (2.782)	5.510 (5.718)	0.047 (0.02)
commentCount	Ridge_TabularOnly_a0.01	3.476 (3.577)	6.237 (6.375)	0.046 (0.01)

Enhanced Model performance over test data:

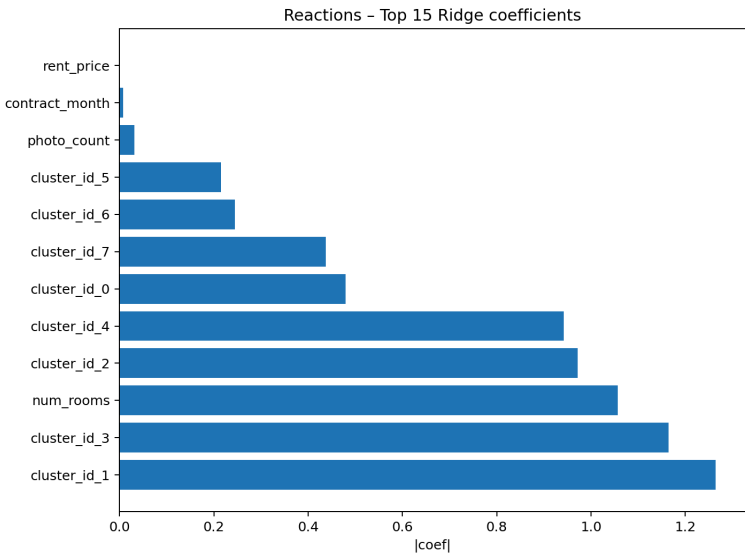
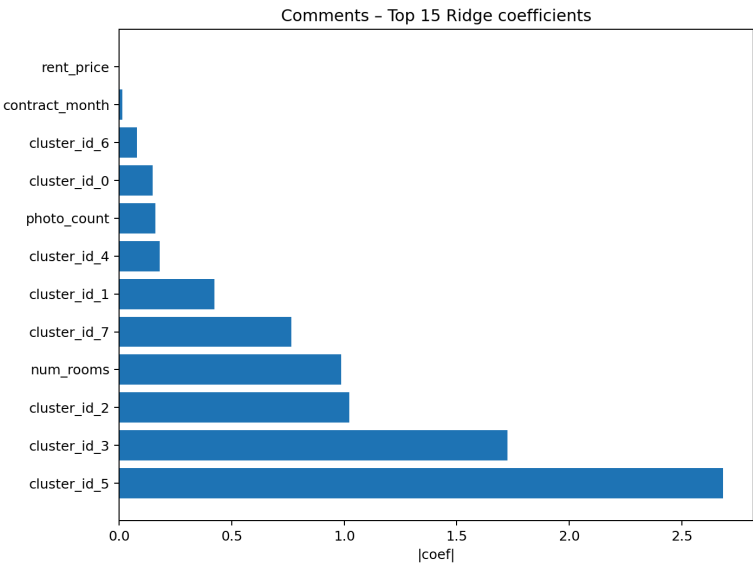


Despite small improvements over the baseline model, our ability to generalize remains low. On the held-out test set we obtain $R^2 \approx 0.047$ for reactions and $R^2 \approx 0.046$ for comments, which means about 95% of the variance remains unexplained. RMSE improves only modestly, roughly 2–4% versus the baseline. Feature analysis also indicates that cluster indicators dominate, while non-cluster tabular features have very small coefficients, so the available signals are weak.

To interpret the Ridge model we simply look at the largest coefficients. To visualize that easily we selected a bar chart of the top 15 coefficients and their value. The coefficient plots show a consistent story across both targets (reactions and comments):

- **Clusters dominate:** cluster indicators carry most of the signal for the two targets; specific clusters differ by target (e.g., strong effects for `cluster_id` ∈ {5, 3} in comments, `cluster_id` ∈ {1, 3} in reactions).
- **Photos Count:** small but positive lift, clearer for reactions/comments, weak for shares.
- **Price:** `rent_price` has a modest positive association.
- **Timing:** `contract_month` is negligible overall.
- **Size:** `num_rooms` skews slightly negative (smaller places draw more attention).

Most Important Parameters in each model:



To get a better sense of whether the issue is with the model's complexity or with the quality of the features, we also experimented with several other models - first with decision trees, and afterwards with XGBoost for the tabular data. Even with the more intricate models, we have not seen a significant improvement over the test data. This indicates that the issues lie in the quality of the engineered features - either they are not as important as we have expected, or there are better ways to process the data (specifically the semantic data), then the simplistic, interoperability-centered methods we used.

Impediments

- Duplicate posts - after the first analysis of the data, we noticed that many posts may appear in both groups, or are reposted word-by-word on various occasions. In order for analysis to be effective, we decided to remove the duplicates based solely on post content. That is, if two posts contained exactly the same text, we removed the latter. This caused us in turn to realize that we need more posts, and we reinitiated the scraping process to form a larger dataset.
- Missing data - We use a simple, consistent policy. Numeric fields are imputed with the column median. `cluster_id` is one-hot encoded to an all-zero dummy vector, effectively an "Unknown" reference category, so missing clusters contribute zero cluster effect and rely on the intercept and other features. Rows with missing labels are removed for that target only.
- Multiple addresses during extraction - as we've mentioned earlier, several posts had multiple addresses, making it more difficult for us to find the one most relevant for analysis. After a few iterations of validating the predictions, we eventually came up with the following rule-based prediction:
 - a. Prefer neighborhoods names over street names
 - b. Select an exact match with a 95-100 confidence score.
 - c. If not exact match, select fuzzy match with 90-100 confidence score and 95-100 similarity score
 - d. Do not select overly-common options, appearing in our blacklist.
- Clustering in high dimensional space - using KMeans and DBSCAN, which we both considered in the beginning, was not feasible in a 768-dimensional space. Thus, we decided to use PCA to reduce the dimension of the embedding space.
- Label imbalance - As we've seen in the EDA process for the milestone, the engagement labels are not balanced. Most posts have little to no engagement, while relatively few have tens or hundreds of likes. This is not an error in the data, but rather part of the pattern we are trying to study. We did not have any solution to this problem, aside from increasing the number of posts we scraped.

Future Work

As could be seen in the results, our models did not perform as good as we had expected. This could have several explanations:

1. The pattern, or manifold, that we are trying to learn is in a higher dimension than our simplistic models could learn. As we've learned in class, natural language is high-dimensional and analysing it using simple models often fails to produce effective results. It is also possible that the fault in the textual/semantic analysis lies in the quality of the embedding of the posts.
2. Another possibility is that some of the important information relevant for this mechanism was not incorporated in our data. One obvious thing, mentioned before, is the images - people react to apartments that are aesthetically appealing. Additionally, people may react differently to the posts based on the profile picture of the person who posted it.

We suggest two possible extensions for this project:

1. Incorporating more sophisticated models - RNN architecture or transformers, to capture the intricate semantics of each post.
2. Analyzing the posts while including the pictures of each post - maybe using an aesthetic score for each picture, or by classifying the contents of the pictures.

Conclusion

Overall, the large efforts we made in order to analyze the causes for post engagement were not as successful as we had hoped, and we only managed to find some smaller contributions to the posts' success. We did find some indicators of good posts in these groups - the more photos you post the better, and it seemed that female-oriented posts and also student-oriented posts had more success than others.

While our ability to predict is not as good as we had hoped for, we did **gain several insights** about the nature of the **apartment market** in Jerusalem - both its location, its typical price range, and the many demographics and target audiences that are part of it. To conclude, we learned a lot about the nature of the data and our problem, though our suggested methodology and solution were less effective than we had hoped.